

Assignment

November 21, 2025

```
[1]: %reload_ext autoreload
      %autoreload 2
```

```
[2]: import torch
      import torch.nn as nn
      from torch.nn import functional as F
      import math
      from torch.nn import LayerNorm
```

```
[3]: from kret_studies import *
      from kret_studies.notebook import *
      from kret_studies.complex import *
```

Loaded environment variables from
/Users/Akseldkw/Desktop/Columbia/COMS4776/.env.

INFO:datasets:JAX version 0.7.2 available.

```
[4]: from kret_studies.kret_torch.mixin.constants import DEVICE_TORCH_STR, \
      ↪DEVICE_LITERAL
```

```
device: DEVICE_LITERAL = DEVICE_TORCH_STR
NUM_EPOCHS = 1
```

```
[5]: from trainer import save_model_auto, load_model_auto, model_filename
```

```
[6]: # !pip install thop
      # !pip install tqdm
```

1 Part 1: Attention and Encoder-Decoder Models

In this section, you'll implement two forms of attention and use them in an encoder-decoder style transformer. You'll then compare performance based on changing dataset size and hidden dimension.

1.1 Step 1: Scaled Dot Product Attention

Implement both the base `scaled_dot_attention` method as well as the `forward` method for the Attention module.

```
[7]: def scaled_dot_attention(
    q: torch.Tensor, k: torch.Tensor, v: torch.Tensor, mask: torch.Tensor | None
    ↪ None = None
):
    """Computes scaled dot product attention with an optional mask.
    q : shape ( batch, k, hidden_size)
    k : shape ( batch, seq_len, hidden_size)
    v : shape ( batch, seq_len, hidden_size)
    mask: optional. shape (k, seq_len)

    returns:
        context : shape (batch, k, hidden_size)
        attention : shape (batch, k, seq_len)
    """

    # -----
    # FILL THIS IN
    # -----
    unnorm_attn = torch.matmul(q, k.transpose(-2, -1)) / math.sqrt(q.size(-1))
    if mask is not None:
        unnorm_attn = unnorm_attn.masked_fill(mask == 0, float("-inf"))
    attention = F.softmax(unnorm_attn, dim=-1)
    context = torch.matmul(attention, v)
    return context, attention
```

```
[8]: class Attention(nn.Module):
    def __init__(self, hidden_size: int):
        super().__init__()
        self.Q = nn.Linear(hidden_size, hidden_size, bias=False)
        self.K = nn.Linear(hidden_size, hidden_size, bias=False)
        self.V = nn.Linear(hidden_size, hidden_size, bias=False)
        self.hidden_size = hidden_size
        self.to(device)
        self.filename = model_filename(hidden_size=hidden_size)

    def forward(self, x: torch.Tensor, annots: torch.Tensor):
        # x : shape ( batch, k, hidden_size)
        # annots : shape ( batch, seq_len, hidden_size)

        # -----
        # FILL THIS IN
        # Pay attention to which inputs you use for the queries, keys, and values.
        ↪ values.
        # -----
        # q =
        # k =
        # v =
```

```

q = self.Q(x)
k = self.K(annots)
v = self.V(annots)
return scaled_dot_attention(q, k, v)

```

1.2 Step 2: Causal Scaled Dot Product Attention

Using your `scaled_dot_attention` method, implement the `forward` method for the `CausalAttention` module.

```

[9]: class CausalAttention(Attention):
    def __init__(self, hidden_size: int):
        super().__init__(hidden_size)
        self.to(device)
        self.filename = model_filename(hidden_size=hidden_size)

    def forward(self, x: torch.Tensor):
        # x : shape ( batch, seq_len, hidden_size)

        # -----
        # FILL THIS IN
        # This should be very similar to your other attention implementation,
        # with the exception that you only have one set of outputs and you
        # must construct a causal attention mask.
        # -----
        # q =
        # k =
        # v =
        # mask =
        seq_len = x.size(1)
        q = self.Q(x)
        k = self.K(x)
        v = self.V(x)
        mask = torch.tril(torch.ones((seq_len, seq_len), device=x.device)).
        ↪ bool()
        return scaled_dot_attention(q, k, v, mask)

```

Simple test cases

Passing this DOES NOT mean your implementation is correct. But, if this fails, then you definitely did something wrong

```

[10]: cros_attn = Attention(32)
      caus_attn = CausalAttention(32)
      caus_attn.Q = cros_attn.Q
      caus_attn.K = cros_attn.K
      caus_attn.V = cros_attn.V

```

```

inp = torch.randn(4, 8, 32).to(device)
inp2 = torch.randn(4, 16, 32).to(device)

print("Checking consistent shape")
assert cros_attn(inp, inp2)[0].shape == inp.shape
assert caus_attn(inp)[0].shape == inp.shape
assert cros_attn(inp, inp2)[1].shape == (4, 8, 16)
assert caus_attn(inp)[1].shape == (4, 8, 8)

print("Checking attention mask")
assert torch.all(caus_attn(inp)[0][:, -1] == cros_attn(inp, inp)[0][:, -1])

print("Passed")

```

```

Checking consistent shape
Checking attention mask
Passed

```

Read through the following code. Make sure you understand what is going on. This code sets up the encoder-decoder architecture. It requires no additional code.

```

[11]: class PositionalEncoding(nn.Module):
    pe: torch.Tensor

    def __init__(self, hidden_size: int, max_len: int = 1000):
        super().__init__()

        pos = torch.arange(max_len).float().unsqueeze(1)
        dim = torch.arange(hidden_size // 2).float().unsqueeze(0)

        div_term = torch.exp(-math.log(10000.0) * (2 * dim) / hidden_size)
        angle = pos * div_term

        pe = torch.zeros(max_len, hidden_size)
        pe[:, 0::2] = torch.sin(angle)
        pe[:, 1::2] = torch.cos(angle)

        self.register_buffer("pe", pe)
        self.to(device)
        self.filename = model_filename(hidden_size=hidden_size, max_len=max_len)

    def forward(self, idx: int):
        return self.pe[idx]

```

```

[12]: class MLP(nn.Module):

    def __init__(self, hidden_size: int):

```

```

        super().__init__()
        self.layer1 = nn.Linear(hidden_size, hidden_size, bias=False)
        self.layer2 = nn.Linear(hidden_size, hidden_size, bias=False)
        self.relu = nn.ReLU()
        self.to(device)
        self.filename = model_filename(hidden_size=hidden_size)

    def forward(self, x: torch.Tensor):
        x = self.layer1(x)
        x = self.relu(x)
        x = self.layer2(x)
        return x

class EncoderBlock(nn.Module):
    def __init__(self, hidden_size: int):
        super().__init__()
        self.norm1 = LayerNorm(hidden_size)
        self.norm2 = LayerNorm(hidden_size)
        self.attn = Attention(hidden_size)
        self.mlp = MLP(hidden_size)
        self.to(device)
        self.filename = model_filename(hidden_size=hidden_size)

    def forward(self, x: torch.Tensor):
        context, attention = self.attn(self.norm1(x), self.norm1(x))
        x = x + context
        x = x + self.mlp(self.norm2(x))
        return x, attention

class DecoderBlock(nn.Module):
    def __init__(self, hidden_size: int):
        super().__init__()
        self.norm1 = LayerNorm(hidden_size)
        self.norm2 = LayerNorm(hidden_size)
        self.cros_attn = Attention(hidden_size)
        self.self_attn = CausalAttention(hidden_size)
        self.mlp = MLP(hidden_size)
        self.to(device)
        self.filename = model_filename(hidden_size=hidden_size)

    def forward(self, x: torch.Tensor, annotation: torch.Tensor):
        context, self_attn = self.self_attn(self.norm1(x))
        x = x + context
        context, cros_attn = self.cros_attn(self.norm2(x), annotation)
        x = x + context

```

```

x = x + self.mlp(self.norm2(x))
return x, self_attn, cross_attn

```

```

[47]: class TransformerEncoderDecoder(nn.Module):
    def __init__(
        self, vocab_size: int, hidden_size: int, num_layers: int, dataset_name:
        ↪str
    ):
        super().__init__()

        self.tok_emb = nn.Embedding(vocab_size, hidden_size)
        self.pos_emb = PositionalEncoding(hidden_size)

        self.encoder_blocks = nn.ModuleList(
            [EncoderBlock(hidden_size) for i in range(num_layers)]
        )
        self.decoder_blocks = nn.ModuleList(
            [DecoderBlock(hidden_size) for i in range(num_layers)]
        )

        self.head = nn.Linear(hidden_size, vocab_size, bias=False)
        self.head.weight = self.tok_emb.weight
        # weight tie, the final layer reuses the weights of the embedding layers
        # this reduces the total number of parameters

        self.apply(self._init_weights) # initialize the weights to a small
        ↪number

        self.to(device)
        self.filename = model_filename(
            hidden_size=hidden_size,
            num_layers=num_layers,
            vocab_size=vocab_size,
            dataset_name=dataset_name,
        )

    def forward(self, inputs: torch.Tensor, annots: torch.Tensor):

        pos_x = torch.arange(
            0, inputs.shape[-1], device=inputs.device
        ).long() # creates tensor([0, 1, 2, 3, ...])
        pos_y = torch.arange(
            0, annots.shape[-1], device=inputs.device
        ).long() # creates tensor([0, 1, 2, 3, ...])

        x = self.tok_emb(inputs) + self.pos_emb(pos_x)
        y = self.tok_emb(annots) + self.pos_emb(pos_y)

```

```

self_attn = []
cros_attn = []

for encode in self.encoder_blocks:
    y, _ = encode(y)

for decode in self.decoder_blocks:
    x, s, c = decode(x, y)
    cros_attn.append(c)
    self_attn.append(s)

return self.head(x), {"self_attn": self_attn, "cros_attn": cros_attn}

def _init_weights(self, module: torch.nn.Module):
    if isinstance(module, nn.Linear):
        torch.nn.init.normal_(module.weight, mean=0.0, std=0.02)
        if module.bias is not None:
            torch.nn.init.zeros_(module.bias)
    elif isinstance(module, nn.Embedding):
        torch.nn.init.normal_(module.weight, mean=0.0, std=0.02)

```

1.3 Step 3: Training the model

You'll now train a set of example models. This section requires no additional code from you, with the exception of setting the datapath. However, **you are required to analyze and include some of its output.**

```

[14]: from dataset import EncoderDecoderDataset, DecoderDataset
      from utils import save_loss_comparison_by_dataset, □
      ↪ save_loss_comparison_by_hidden
      from torch.utils.data import DataLoader
      from trainer import train
      import torch.optim as optim

data_path = str("./") # Change data_path to where the data is stored

encdec_ds_small = EncoderDecoderDataset(data_path, "pig_latin_small")
encdec_ds_large = EncoderDecoderDataset(data_path, "pig_latin_large")

[15]: print("Dataset sample:")
      print("Input:", encdec_ds_small.to_token(encdec_ds_small[0][0].tolist()))
      print("Annotation:", encdec_ds_small.to_token(encdec_ds_small[0][1].tolist()))
      print("Target:", encdec_ds_small.to_token(encdec_ds_small[0][2].tolist()))

```

```

Dataset sample:
Input: <SOS>anxiouslyway
Annotation: anxiously

```

Target: anxiouslyway<EOS>

```
[55]: model = load_model_auto(
    TransformerEncoderDecoder,
    vocab_size=len(encdec_ds_small.idx_to_token),
    hidden_size=16,
    num_layers=4,
    dataset_name="small",
)
```

[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/TransformerEncoderDecoder/Model_ds_namesmall_vocab30_hid16_layers4.pt onto mps device.

[WARN] Load from disk failed, keeping random initialization... [Errno 2] No such file or directory: '/Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/TransformerEncoderDecoder/Model_ds_namesmall_vocab30_hid16_layers4.pt'

```
[56]: optimizer = optim.AdamW(model.parameters(), lr=1e-2)
scheduler = optim.lr_scheduler.ExponentialLR(optimizer, gamma=0.95)

results = train(
    model=model,
    dataset=encdec_ds_small,
    num_epochs=NUM_EPOCHS,
    optimizer=optimizer,
    scheduler=scheduler,
    device=device,
    batch_size=32,
)
save_model_auto(model)
```

Epoch 1 | Train loss: 2.608985 | Val loss 2.313728

```
[56]: PosixPath('/Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/TransformerEncoderDecoder/Model_ds_namesmall_vocab30_hid16_layers4.pt')
```

```
[57]: def encdec_generate(
    input: str,
    model: TransformerEncoderDecoder,
    dataset: EncoderDecoderDataset,
    max_len: int = 20,
):
    gen_string = ""
    idx = torch.tensor([[dataset.start_idx]]).to(device)
    annots = torch.tensor(
        [dataset.to_idx(input) + [dataset.end_idx] * (max(0, 12 - len(input)))]
    ).to(device)
```



```
# pad the input string with <EOS> since the collate_fn had to pad the
↪strings
```

```
for i in range(max_len):
    logits = model(idx, annots)[0]

    token_idx: torch.Tensor = logits[0, -1:].argmax(dim=-1)
    idx = torch.cat([idx, token_idx.unsqueeze(0)], dim=1)

    if token_idx.item() == dataset.end_idx:
        break
    else:
        gen_string = gen_string + dataset.idx_to_token[int(token_idx.
↪item())]
    return gen_string

def encdec_translate(
    sentence: str, model: TransformerEncoderDecoder, dataset: ↪
↪EncoderDecoderDataset
):
    words = sentence.split()
    return " ".join([encdec_generate(w, model, dataset) for w in words])
```

```
[52]: encdec_translate("the", model, encdec_ds_small)
```

```
[52]: 'aaaaay'
```

```
[53]: encdec_translate("the air conditioning is working", model, encdec_ds_small)
```

```
[53]: 'aaaaay aaaaay enneeeeeeeaaay aaaaay aeaaaaay'
```

```
[54]: encdec_translate("testing some random words", model, encdec_ds_small)
```

```
[54]: 'aeaaaaay aaaaay aeaaaaay assaaay'
```

```
[59]: configs = [
    (16, "small", encdec_ds_small),
    (16, "large", encdec_ds_large),
    (32, "small", encdec_ds_small),
    (32, "large", encdec_ds_large),
]

all_results = {}

for hidden_size, dataset_name, dataset in configs:
```

```

print(f"\nTraining model: hidden_size={hidden_size},  

dataset={dataset_name}")

model = load_model_auto(
    TransformerEncoderDecoder,
    vocab_size=len(dataset.idx_to_token),
    hidden_size=hidden_size,
    num_layers=4,
    dataset_name=dataset_name,
)

optimizer = optim.AdamW(model.parameters(), lr=1e-2)
scheduler = optim.lr_scheduler.ExponentialLR(optimizer, gamma=0.95)

results = train(
    model=model,
    dataset=dataset,
    num_epochs=NUM_EPOCHS,
    optimizer=optimizer,
    scheduler=scheduler,
    device=device,
    batch_size=32 if dataset_name == "small" else 256,
)
all_results[(hidden_size, dataset_name)] = results
save_model_auto(model)

save_loss_comparison_by_dataset(all_results)
save_loss_comparison_by_hidden(all_results)

```

```

Training model: hidden_size=16, dataset=small
[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-
Data/data/homework/HW2/TransformerEncoderDecoder/Model_ds_namesmall_vocab30_hid1
6_layers4.pt onto mps device.
Epoch 1 | Train loss: 1.833445 | Val loss 1.650882

```

```

Training model: hidden_size=16, dataset=large
[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-
Data/data/homework/HW2/TransformerEncoderDecoder/Model_ds_namelarge_vocab30_hid1
6_layers4.pt onto mps device.
Epoch 1 | Train loss: 2.117964 | Val loss 1.871782

```

```

Training model: hidden_size=32, dataset=small
[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-
Data/data/homework/HW2/TransformerEncoderDecoder/Model_ds_namesmall_vocab30_hid3
2_layers4.pt onto mps device.
Epoch 1 | Train loss: 2.147391 | Val loss 1.942184

```

```

Training model: hidden_size=32, dataset=large
[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-
Data/data/homework/HW2/TransformerEncoderDecoder/Model_ds_name_large_vocab30_hid3
2_layers4.pt onto mps device.
Epoch 1 | Train loss: 2.169914 | Val loss 1.921395

<Figure size 640x480 with 0 Axes>

<Figure size 640x480 with 0 Axes>

```

2 Part 2: Decoder-only Transformers and Multi-Head Attention

In this section, you'll train transformers that only use a decoder instead of a dual encoder-decoder architecture. You'll also implement and experiment with multi-head attention. Finally, you'll compare increasing parameter count with multi-head attention vs. increasing the parameter count with a larger hidden dimensionality.

2.1 Step 1: Implement the Decoder-Only Forward Method

```

[ ]: class DecoderOnlyBlock(nn.Module):
    def __init__(self, hidden_size: int):
        super().__init__()
        self.norm1 = LayerNorm(hidden_size)
        self.norm2 = LayerNorm(hidden_size)
        self.attn = CausalAttention(hidden_size)
        self.mlp = MLP(hidden_size)
        self.to(device)
        self.filename = model_filename(hidden_size=hidden_size)

    def forward(self, x: torch.Tensor):
        context, attention = self.attn(self.norm1(x))
        x = x + context
        x = x + self.mlp(self.norm2(x))
        return x, attention

class DecoderTransformer(nn.Module):
    def __init__(self, vocab_size: int, hidden_size: int, num_layers: int):
        super().__init__()

        self.tok_emb = nn.Embedding(vocab_size, hidden_size)
        self.pos_emb = PositionalEncoding(hidden_size)

        self.blocks = nn.ModuleList(
            [DecoderOnlyBlock(hidden_size) for i in range(num_layers)]
        )
        self.head = nn.Linear(hidden_size, vocab_size, bias=False)
        self.head.weight = self.tok_emb.weight

```

```

        # weight tie, the final layer reuses the weights of the embedding layers
        # this reduces the total number of parameters
        self.to(device)
        self.filename = model_filename(
            hidden_size=hidden_size, num_layers=num_layers,
            vocab_size=vocab_size
        )

    def forward(self, inputs: torch.Tensor):
        """Forward pass of the Transformer decoder.

        Arguments:
            inputs: Input token indexes across a batch for all the time step.
            (batch_size x decoder_seq_len)
        Returns:
            output: Un-normalized scores for each token in the vocabulary,
            across a batch for all the decoding time steps. (batch_size x
            decoder_seq_len x vocab_size)
            attentions: The stacked attention weights applied to the encoder
            annotations (batch_size x encoder_seq_len x decoder_seq_len)
        """
        # -----
        # FILL THIS IN
        # -----
        # pos =
        # x =
        # self_attn = []

        # for block in self.blocks:
        #     ...
        #     #
        #     # unnormalized_scores =
        pos = torch.arange(0, inputs.shape[-1], device=inputs.device).long()
        x = self.tok_emb(inputs) + self.pos_emb(pos)
        self_attn = []
        for block in self.blocks:
            x, s = block(x)
            self_attn.append(s)
        unnormalized_scores = self.head(x)

        return unnormalized_scores, {"self_attn": self_attn}

```

```

[25]: decoder_ds_small = DecoderDataset(data_path, "pig_latin_small")
      decoder_ds_large = DecoderDataset(data_path, "pig_latin_large")

```

```
[26]: print("Dataset sample:")
print("Input:", decoder_ds_small.to_token(decoder_ds_small[0][0].tolist()))
print("Target:", decoder_ds_small.to_token(decoder_ds_small[0][1].tolist()))
```

Dataset sample:

Input: <SOS>anxiously<EOP>anxiouslyway

Target: anxiously<EOP>anxiouslyway<EOS>

2.2 Step 2: Format for decoder-only generation.

Implement the missing line in `decoder_generate` to create an input string that looks like:

<SOS> input sequence <EOP>

You should use the `dataset.start_idx`, `dataset.eop_idx` attributes and the `dataset.to_idx()` method.

```
[27]: def decoder_generate(
    input: str,
    model: DecoderTransformer,
    dataset: DecoderDataset,
    max_len: int = 20,
):
    gen_string = ""
    # idx = TODO
    idx = dataset.to_idx(input)
    idx = torch.tensor([idx]).to(device)

    for i in range(max_len):
        logits = model(idx)[0]

        token_idx = logits[0, -1:].argmax(dim=-1)
        idx = torch.cat([idx, token_idx.unsqueeze(0)], dim=1)

        if token_idx.item() == dataset.end_idx:
            break
        else:
            gen_string = gen_string + dataset.idx_to_token[token_idx.item()]
    return gen_string

def decoder_translate(
    sentence: str, model: DecoderTransformer, dataset: DecoderDataset
):
    words = sentence.split()
    return " ".join([decoder_generate(w, model, dataset) for w in words])
```

2.3 Step 3: Train the model

We'll now train a decoder only model and test it out on an example. Use the output of this to guide your answer for question 3.

```
[28]: model = load_model_auto(
    DecoderTransformer,
    vocab_size=len(decoder_ds_small.idx_to_token),
    hidden_size=16,
    num_layers=4,
)
```

```
[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-
Data/data/homework/HW2/DecoderTransformer/Model_vocab30_hid16_layers4.pt onto
mps device.
```

```
[WARN] Load from disk failed, keeping random initialization... [Errno 2] No such
file or directory: '/Users/Akseldkw/coding/Columbia/COMS4776-
Data/data/homework/HW2/DecoderTransformer/Model_vocab30_hid16_layers4.pt'
```

```
[ ]: optimizer = optim.AdamW(model.parameters(), lr=1e-2)
scheduler = optim.lr_scheduler.ExponentialLR(optimizer, gamma=0.95)

results = train(
    model=model,
    dataset=decoder_ds_small,
    num_epochs=NUM_EPOCHS,
    optimizer=optimizer,
    scheduler=scheduler,
    device=device,
    batch_size=32,
)
save_model_auto(model)
```

```
Epoch 1 | Train loss: 2.981663 | Val loss 1.899971
```

```
[ ]: PosixPath('/Users/Akseldkw/coding/Columbia/COMS4776-
Data/data/homework/HW2/DecoderTransformer/Model_vocab30_hid16_layers4.pt')
```

```
[30]: decoder_translate("the air conditioning is working", model, decoder_ds_small)
```

```
[30]: 'ay araranararay ngsingsay sssssssssssssssssss ngngrrrringray'
```

2.4 Step 4: Implement Multi-Head Attention

In this next step, you'll implement multi-head attention. In this setting, we'll have the dimensionality for each of the heads be $(\text{total hidden dimensionality})/(\text{number of heads})$. We'll use a projection head to combine the outputs of these different attention heads after concatenating their outputs together (which can also be achieved using tensor reshaping).

```
[31]: # TO DO:

def multihead_scaled_dot_attention(
    q: torch.Tensor, k: torch.Tensor, v: torch.Tensor, mask: torch.Tensor |
    ↪None = None
):
    """Compute multi-head dot product attention.
        q : shape (batch, num_heads, k, hidden_size)
        k : shape (batch, num_heads, seq_len, hidden_size)
        v : shape (batch, num_heads, seq_len, hidden_size)

    returns:
        context : shape (batch, num_heads, k, hidden_size)
        attention : shape (batch, num_heads, k, seq_len)
    """
    # Hint: if you implemented scaled_dot_attention well, the code for this ↪
    ↪should be the exact same.

    # -----
    # FILL THIS IN
    # -----
    # unnorm_attn =
    # masked_unnorm_attn =
    # attention =
    # context =
    unnorm_attn = torch.matmul(q, k.transpose(-2, -1)) / math.sqrt(q.size(-1))
    if mask is not None:
        unnorm_attn = unnorm_attn.masked_fill(mask == 0, float("-inf"))
    attention = F.softmax(unnorm_attn, dim=-1)
    context = torch.matmul(attention, v)
    return context, attention
```

```
[ ]: class MultiheadCausalAttention(Attention):
    proj: nn.Linear

    def __init__(self, hidden_size: int, num_heads: int):
        super().__init__(hidden_size)
        # self.Q, self.K, self.V already defined as nn.Linear(hidden_size, ↪
    ↪hidden_size)
        self.num_heads = num_heads
        # -----
        # FILL THIS IN
        # -----
        # if self.num_heads > 1:
        #     self.proj = nn.Linear( , , bias=False)
        if self.num_heads > 1:
```

```

        self.proj = nn.Linear(hidden_size, hidden_size, bias=False)
        self.to(device)
        self.filename = model_filename(hidden_size=hidden_size,
        ↪ num_heads=num_heads)

    def forward(self, x: torch.Tensor):
        """Compute causal multi-head dot product attention from an input
        ↪ sequence.
            x : shape ( batch, seq_len, hidden_size)
            k : shape (batch, num_heads, seq_len, hidden_size)
            v : shape (batch, num_heads, seq_len, hidden_size)

        returns:
            context : shape ( batch, seq_len, hidden_size)
            attention : shape (batch, num_heads, k, seq_len)
        """
        # -----
        # FILL THIS IN
        # -----
        # unnorm_attn =
        # masked_unnorm_attn =
        # attention =
        # context =

        # B, T, C = x.size()
        # q = self.Q(x).view( , , , ).transpose(1, 2)
        # k = self.K(x).view( , , , ).transpose(1, 2)
        # v = self.V(x).view( , , , ).transpose(1, 2)

        # mask =

        # context, attention = multihead_scaled_dot_attention(q, k, v, mask)

        # Hint: you may find the .transpose(), .contiguous(), and .view()
        # methods helpful for creating the reshaped context.

        # reshaped_context =

        # Only perform a projection if the number of heads is more than 1.
        B, T, C = x.size()
        q = self.Q(x).view(B, T, self.num_heads, C // self.num_heads).
        ↪ transpose(1, 2)
        k = self.K(x).view(B, T, self.num_heads, C // self.num_heads).
        ↪ transpose(1, 2)
        v = self.V(x).view(B, T, self.num_heads, C // self.num_heads).
        ↪ transpose(1, 2)
        mask = torch.tril(torch.ones((T, T), device=x.device)).bool()

```



```

context, attention = multihead_scaled_dot_attention(q, k, v, mask)
reshaped_context = context.transpose(1, 2).contiguous().view(B, T, C)
if self.num_heads > 1:

    return self.proj(reshaped_context), attention

else:

    return reshaped_context, attention

```

```

[ ]: class MultiheadDecoderBlock(DecoderOnlyBlock):
    def __init__(self, hidden_size: int, num_heads: int):
        super().__init__(hidden_size)
        self.attn = MultiheadCausalAttention(hidden_size, num_heads)
        self.to(device)
        self.filename = model_filename(hidden_size=hidden_size,
↪num_heads=num_heads)

class MultiheadDecoderTransformer(DecoderTransformer):
    def __init__(
        self, vocab_size: int, hidden_size: int, num_layers: int, num_heads: int
    ):
        super().__init__(vocab_size, hidden_size, num_layers)
        self.num_heads = num_heads
        self.num_layers = num_layers
        self.vocab_size = vocab_size
        self.hidden_size = hidden_size

        self.blocks = nn.ModuleList(
            [MultiheadDecoderBlock(hidden_size, num_heads) for i in
↪range(num_layers)]
        )
        self.to(device)
        self.filename = model_filename(
            hidden_size=hidden_size,
            num_layers=num_layers,
            vocab_size=vocab_size,
            num_heads=num_heads,
        )

```

2.5 Step 5: Compare Parameter Usage Methods

Here, you'll train 5 models with varying numbers of attention heads and hidden dimensionalities. You'll compare their parameter efficiency. Start by implementing the `calc_parameters_decoder` method, and then run the code to train the models.

```
[34]: def calc_parameters_decoder(model: MultiheadDecoderTransformer) -> int:
    # -----
    # FILL THIS IN
    # -----
    # params = TODO # embedding
    # params += TODO # QKV linear projections
    # if model.num_heads > 1:
    #     params += TODO # attention linear projection
    # params += TODO # MLP linear
    # params += TODO # layer norms Hint: LayerNorm(x) uses x parameters

    # Hint: Read the code carefully:
    # params += TODO # head
    params = model.vocab_size * model.hidden_size # embedding
    params += 3 * (model.hidden_size * model.hidden_size) # QKV linear
    ↪projections
    if model.num_heads > 1:
        params += model.hidden_size * model.hidden_size # attention linear
    ↪projection
    params += 2 * (
        4 * model.hidden_size * model.hidden_size
    ) # MLP linear (2 per block)
    params += 2 * 2 * model.hidden_size # layer norms (2 per block)

    return params
```

```
[ ]: import matplotlib.pyplot as plt
import numpy as np

configs = [
    (16, 4, 4, 64), # (hidden_size, num_heads, num_layers, batch_size)
    (16, 2, 4, 64),
    (16, 1, 4, 64),
    (24, 1, 4, 64),
    (32, 1, 4, 64),
]

all_results = {}

for hidden_size, num_heads, num_layers, batch_size in configs:
    print(
        f"\nTraining model: hidden_size={hidden_size}, num_heads={num_heads},
    ↪num_layers={num_layers}"
    )

    model = load_model_auto(
        MultiheadDecoderTransformer,
```

```

        vocab_size=len(decoder_ds_large.idx_to_token),
        hidden_size=hidden_size,
        num_layers=num_layers,
        num_heads=num_heads,
    )

    optimizer = optim.AdamW(model.parameters(), lr=5e-3)
    scheduler = optim.lr_scheduler.ExponentialLR(optimizer, gamma=0.95)

    results = train(
        model=model,
        dataset=decoder_ds_large,
        num_epochs=NUM_EPOCHS,
        optimizer=optimizer,
        scheduler=scheduler,
        device=device,
        batch_size=batch_size,
    )
    save_model_auto(model)

    parameters = calc_parameters_decoder(model)
    x_points = np.linspace(0, len(results["val_loss"]),
        len(results["val_loss"]))
    plt.plot(
        x_points,
        results["val_loss"],
        label="heads = {}, dim={}, params={}".format(
            num_heads, hidden_size, parameters
        ),
    )

plt.xlabel("Epochs")
plt.ylabel("Validation Loss")
plt.title("Compute Optimal Models")
plt.grid()
plt.legend()
plt.show()

```

Training model: hidden_size=16, num_heads=4, num_layers=4

[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid16_layers4_heads4.pt onto mps device.

[WARN] Load from disk failed, keeping random initialization... [Errno 2] No such file or directory: '/Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid16_layers4_heads4.pt'

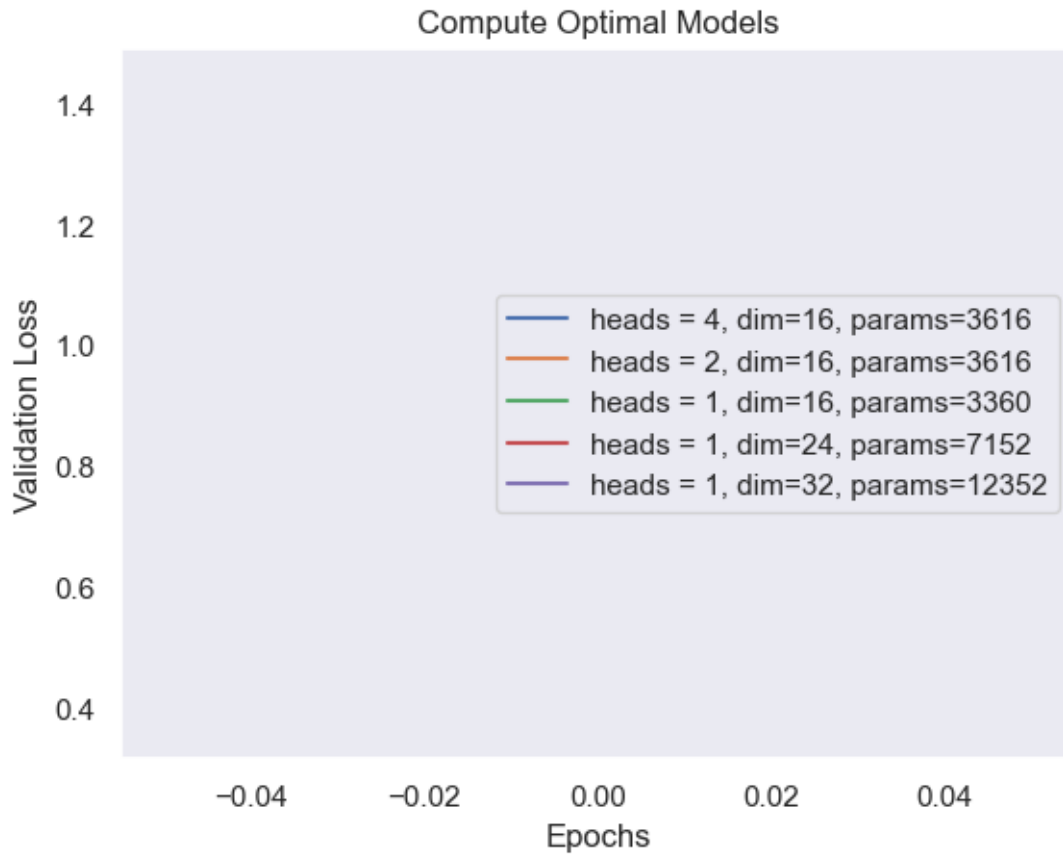
Epoch 1 | Train loss: 2.076174 | Val loss 1.188008

Training model: hidden_size=16, num_heads=2, num_layers=4
[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid16_layers4_heads2.pt onto mps device.
[WARN] Load from disk failed, keeping random initialization... [Errno 2] No such file or directory: '/Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid16_layers4_heads2.pt'
Epoch 1 | Train loss: 2.179173 | Val loss 1.316626

Training model: hidden_size=16, num_heads=1, num_layers=4
[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid16_layers4_heads1.pt onto mps device.
[WARN] Load from disk failed, keeping random initialization... [Errno 2] No such file or directory: '/Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid16_layers4_heads1.pt'
Epoch 1 | Train loss: 2.194134 | Val loss 1.438002

Training model: hidden_size=24, num_heads=1, num_layers=4
[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid24_layers4_heads1.pt onto mps device.
[WARN] Load from disk failed, keeping random initialization... [Errno 2] No such file or directory: '/Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid24_layers4_heads1.pt'
Epoch 1 | Train loss: 1.953525 | Val loss 0.668632

Training model: hidden_size=32, num_heads=1, num_layers=4
[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid32_layers4_heads1.pt onto mps device.
[WARN] Load from disk failed, keeping random initialization... [Errno 2] No such file or directory: '/Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid32_layers4_heads1.pt'
Epoch 1 | Train loss: 1.867597 | Val loss 0.369293



```
[36]: plt.clf()
```

<Figure size 640x480 with 0 Axes>

3 Part 3: IsoFLOP Profiling and Scaling Laws

In this final section, you'll compare different architectures based on their compute efficiency. You will also determine the optimal number of parameters and tokens for a fixed compute budget.

While this section does not require any additional code from you, **use the output of it to guide your written responses.**

```
[37]: from utils import (  
    plot_scaling_law,  
    plot_scaling_law_poly,  
    interpolate,  
    plot_flops_tokens,  
    plot_isoflop,  
    plot_flops_params,  
)
```

```
import numpy as np
```

3.1 Step 1: Train models, Compare FLOPS vs Validation Loss

We'll inspect the compute efficiency of different models by plotting their validation loss against the number of FLOPs used in training up to that point.

```
[38]: from types_hw2 import TrainerHistDict
```

```
[39]: configs = [
    (40, 2, 4, 256), # (hidden_size, num_heads, num_layers, batch_size)
    (32, 2, 4, 128),
    (24, 2, 4, 64),
    (16, 1, 4, 64),
    (16, 1, 3, 64),
    (12, 1, 3, 64),
    (12, 1, 2, 64),
]

all_results: dict[int, TrainerHistDict] = {}

for hidden_size, num_heads, num_layers, batch_size in configs:
    print(
        f"\nTraining model: hidden_size={hidden_size}, num_heads={num_heads},
        ↪num_layers={num_layers}"
    )

    model = load_model_auto(
        MultiheadDecoderTransformer,
        vocab_size=len(decoder_ds_large.idx_to_token),
        hidden_size=hidden_size,
        num_layers=num_layers,
        num_heads=num_heads,
    )

    optimizer = optim.AdamW(model.parameters(), lr=5e-3)
    scheduler = optim.lr_scheduler.ExponentialLR(optimizer, gamma=0.95)

    results = train(
        model=model,
        dataset=decoder_ds_large,
        num_epochs=NUM_EPOCHS,
        optimizer=optimizer,
        scheduler=scheduler,
        device=device,
        batch_size=batch_size,
```

```

)
save_model_auto(model)

parameters = calc_parameters_decoder(model)
all_results[parameters] = results

```

Training model: hidden_size=40, num_heads=2, num_layers=4
 [INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid40_layers4_heads2.pt onto mps device.
 [WARN] Load from disk failed, keeping random initialization... [Errno 2] No such file or directory: '/Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid40_layers4_heads2.pt'
 Epoch 1 | Train loss: 4.492354 | Val loss 1.408214

Training model: hidden_size=32, num_heads=2, num_layers=4
 [INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid32_layers4_heads2.pt onto mps device.
 [WARN] Load from disk failed, keeping random initialization... [Errno 2] No such file or directory: '/Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid32_layers4_heads2.pt'
 Epoch 1 | Train loss: 2.786649 | Val loss 0.966770

Training model: hidden_size=24, num_heads=2, num_layers=4
 [INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid24_layers4_heads2.pt onto mps device.
 [WARN] Load from disk failed, keeping random initialization... [Errno 2] No such file or directory: '/Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid24_layers4_heads2.pt'
 Epoch 1 | Train loss: 1.831188 | Val loss 0.501310

Training model: hidden_size=16, num_heads=1, num_layers=4
 [INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid16_layers4_heads1.pt onto mps device.
 Epoch 1 | Train loss: 1.242564 | Val loss 0.915337

Training model: hidden_size=16, num_heads=1, num_layers=3
 [INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid16_layers3_heads1.pt onto mps device.
 [WARN] Load from disk failed, keeping random initialization... [Errno 2] No such

```
file or directory: '/Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid16_layers3_heads1.pt'
```

```
Epoch 1 | Train loss: 2.222150 | Val loss 1.423268
```

```
Training model: hidden_size=12, num_heads=1, num_layers=3
```

```
[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid12_layers3_heads1.pt onto mps device.
```

```
[WARN] Load from disk failed, keeping random initialization... [Errno 2] No such file or directory: '/Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid12_layers3_heads1.pt'
```

```
Epoch 1 | Train loss: 2.427160 | Val loss 1.731618
```

```
Training model: hidden_size=12, num_heads=1, num_layers=2
```

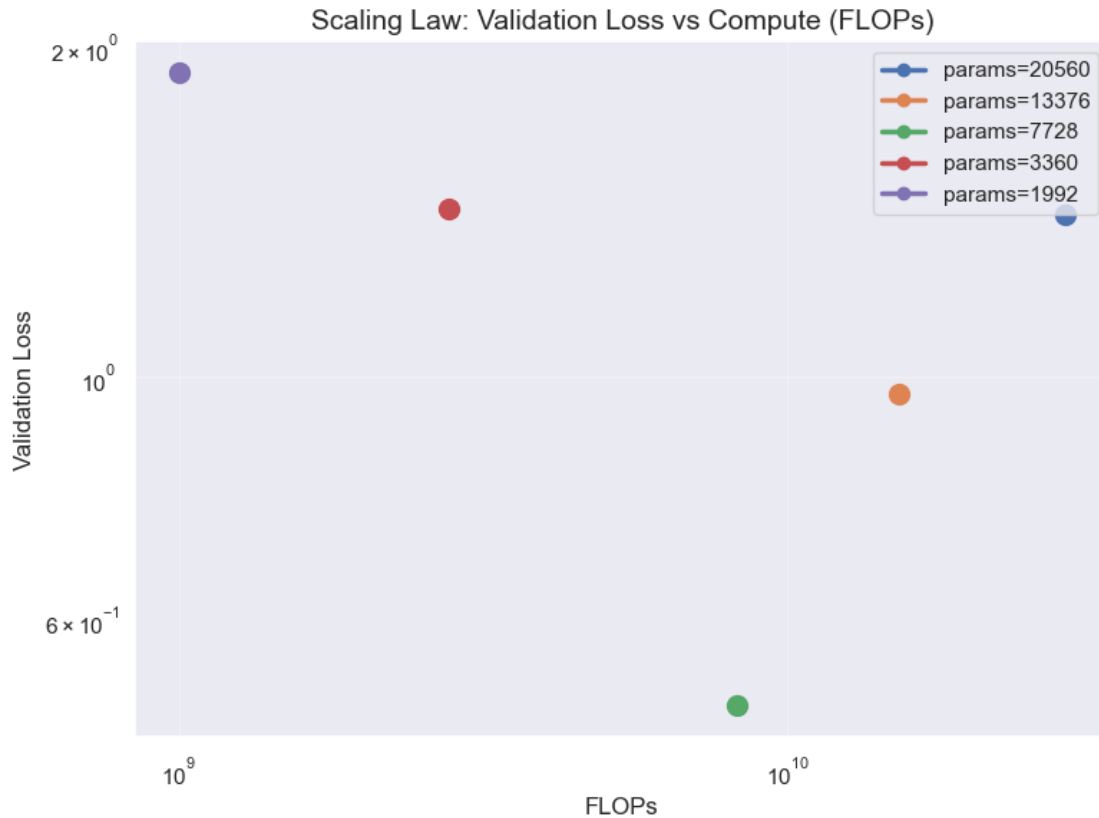
```
[INFO] Loading model weights from /Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid12_layers2_heads1.pt onto mps device.
```

```
[WARN] Load from disk failed, keeping random initialization... [Errno 2] No such file or directory: '/Users/Akseldkw/coding/Columbia/COMS4776-Data/data/homework/HW2/MultiheadDecoderTransformer/Model_vocab30_hid12_layers2_heads1.pt'
```

```
Epoch 1 | Train loss: 2.677001 | Val loss 1.895238
```

```
[40]: plot_scaling_law(all_results)

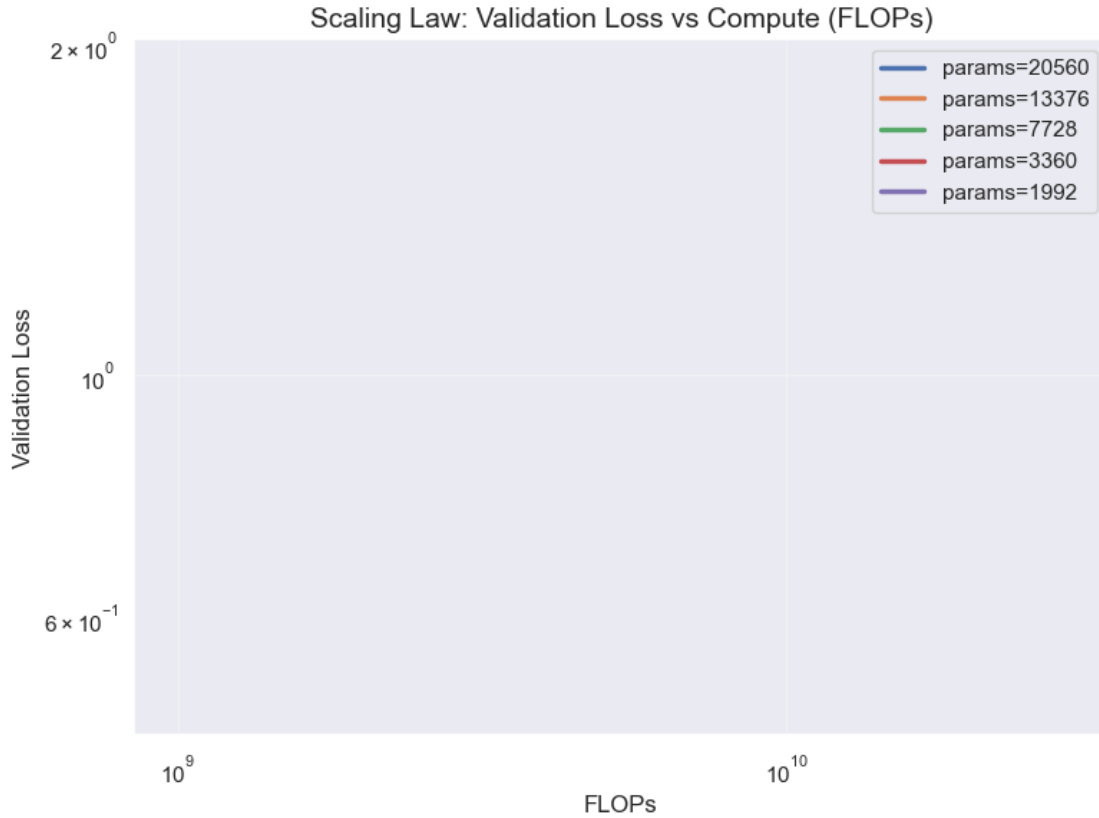
      plot_scaling_law_poly(all_results)
```

```

/Users/Akseldkw/Desktop/Columbia/COMS4776/homework/HW2/utils.py:186:
RankWarning: Polyfit may be poorly conditioned
    poly_coeffs = np.polyfit(np.log10(flops), np.log10(val_loss), 5)
/Users/Akseldkw/Desktop/Columbia/COMS4776/homework/HW2/utils.py:186:
RankWarning: Polyfit may be poorly conditioned
    poly_coeffs = np.polyfit(np.log10(flops), np.log10(val_loss), 5)
/Users/Akseldkw/Desktop/Columbia/COMS4776/homework/HW2/utils.py:186:
RankWarning: Polyfit may be poorly conditioned
    poly_coeffs = np.polyfit(np.log10(flops), np.log10(val_loss), 5)
/Users/Akseldkw/Desktop/Columbia/COMS4776/homework/HW2/utils.py:186:
RankWarning: Polyfit may be poorly conditioned
    poly_coeffs = np.polyfit(np.log10(flops), np.log10(val_loss), 5)
/Users/Akseldkw/Desktop/Columbia/COMS4776/homework/HW2/utils.py:186:
RankWarning: Polyfit may be poorly conditioned
    poly_coeffs = np.polyfit(np.log10(flops), np.log10(val_loss), 5)

```



3.2 Step 2: Optimal Number of Parameters

Using the previously created training runs, we'll fit quadratic functions to the (parameter count, validation loss) pairs to estimate the optimal number of parameters for a given compute budget. We'll then fit a line to this set of optimal parameters to estimate the optimal number of parameters for a compute budget of $1e15$.

```
[41]: def find_optim_params(params: torch.Tensor, loss: torch.Tensor):
    x = np.array(params.cpu())
    y = np.array(loss.cpu())

    p = np.polyfit(np.log10(x), y, 2)
    optimal_log_params = -(p[1]) / (2 * p[0])
    optimal_params = 10 ** (optimal_log_params)

    return optimal_params

target_flops = [8e9, 16e9, 32e9, 64e9, 128e9]
colors = ["#E41A1C", "#377EB8", "#4DAF4A", "#984EA3", "#FF7F00"]
params_vs_loss = []
```

```

optimal_params = []

for target in target_flops:
    parameters = []
    val_loss = []

    for params, results in all_results.items():
        loss = interpolate(results["val_loss"], results["flops"], target, deg=4)

        # interpolate returns None if target is out of range
        if loss != None:
            parameters.append(params)
            val_loss.append(loss)

    sorted_indices = torch.argsort(torch.tensor(parameters)).to(device)
    parameters = torch.tensor(parameters).to(device)[sorted_indices]
    val_loss = torch.tensor(val_loss).to(device)[sorted_indices]
    try:
        optimal_params.append(find_optim_params(parameters, val_loss))
        params_vs_loss.append((parameters.tolist(), val_loss.tolist()))
    except Exception as e:
        print(
            "\n\n{optimal_params=}\n\n{params_vs_loss=}\n\n{parameters=}\n\n{val_loss=}\n\n{target=}"
        )

```

```

/var/folders/dj/k5kk5wk90vvcnb_wphpw490r0000gn/T/ipykernel_91624/2694499551.py:2
: DeprecationWarning: __array__ implementation doesn't accept a copy keyword, so
passing copy=False failed. __array__ must implement 'dtype' and 'copy' keyword
arguments. To learn more, see the migration guide
https://numpy.org/devdocs/numpy\_2\_0\_migration\_guide.html#adapting-to-changes-in-the-copy-keyword

```

```

x = np.array(params.cpu())
/var/folders/dj/k5kk5wk90vvcnb_wphpw490r0000gn/T/ipykernel_91624/2694499551.py:3
: DeprecationWarning: __array__ implementation doesn't accept a copy keyword, so
passing copy=False failed. __array__ must implement 'dtype' and 'copy' keyword
arguments. To learn more, see the migration guide
https://numpy.org/devdocs/numpy\_2\_0\_migration\_guide.html#adapting-to-changes-in-the-copy-keyword
y = np.array(loss.cpu())

```

```

[42]: print(
    type(target_flops),
    type(target_flops[0]),
    type(optimal_params), # type(optimal_params[0]),
    type(all_results),
    type(params_vs_loss), # type(params_vs_loss[0]),

```

```

type(colors),
type(colors[0]),
)

```

```

<class 'list'> <class 'float'> <class 'list'> <class 'dict'> <class 'list'>
<class 'list'> <class 'str'>

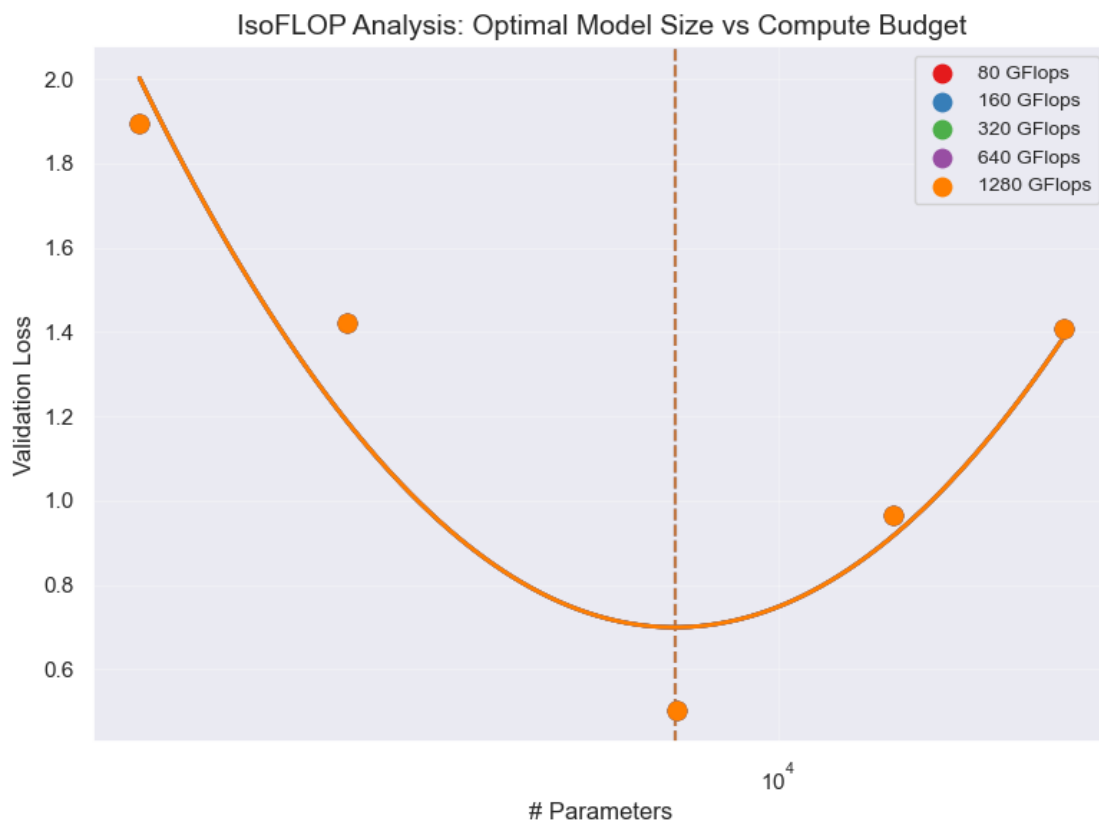
```

```
[43]: plot_isoflop(target_flops, colors, params_vs_loss, optimal_params)
```

```

8.00 GFlops: Optimal params = 7,701
16.00 GFlops: Optimal params = 7,701
32.00 GFlops: Optimal params = 7,701
64.00 GFlops: Optimal params = 7,701
128.00 GFlops: Optimal params = 7,701

```



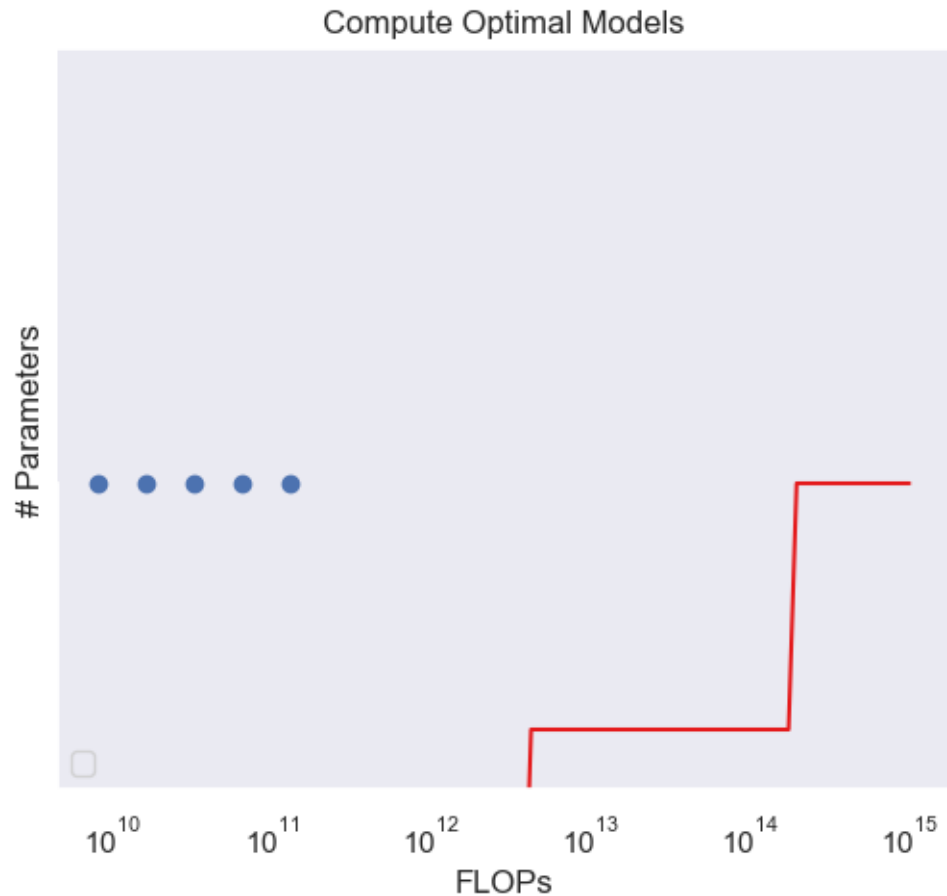
```
[44]: plot_flops_params(target_flops, optimal_params)
```

```
y = 2.644567614964774e-16x + 3.886552322415936
```

```
/Users/Akseldkw/Desktop/Columbia/COMS4776/homework/HW2/utils.py:282:
```

```
UserWarning: No artists with labels found to put in legend. Note that artists
whose label start with an underscore are ignored when legend() is called with no
```

```
argument.  
ax.legend(loc="lower left")
```



3.3 Step 3: Optimal Number of Tokens

Finally, we'll perform a similar analysis to the previous step, only this time determining the optimal number of tokens for a budget of $1e15$ flops.

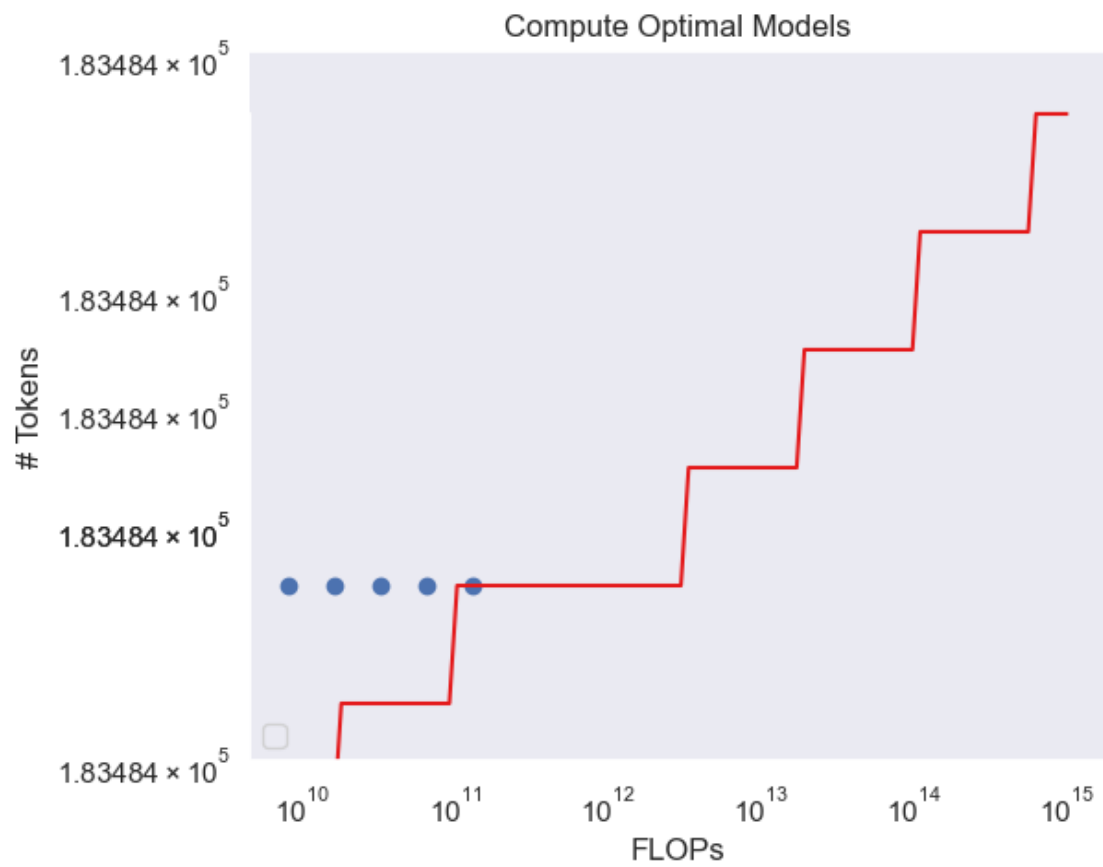
```
[45]: plot_flops_tokens(target_flops, optimal_params, all_results)
```

```
y = 1.1711656580558287e-15x + 5.26359811181344
```

```
/Users/Akseldkw/Desktop/Columbia/COMS4776/homework/HW2/utils.py:323:
```

```
UserWarning: No artists with labels found to put in legend. Note that artists  
whose label start with an underscore are ignored when legend() is called with no  
argument.
```

```
ax.legend(loc="lower left")
```



[]: